BME302- FLUID MECHANICS AND FLUID MACHINES

Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course , the student will be able to		
CO 1	Understand the application of mass and momentum conservation laws for fluid flows.	**K2**
CO 2	Understand the importance of dimensional analysis.	**K2**
CO 3	Evaluate the velocity and pressure variations in various types of simple flows.	**K3**
CO 4	Mathematically analyze the flow in water pumps and turbines.	**K3**
CO 5	Understand about the functioning of centrifugal and reciprocating pumps.	**K2**
DETAILED SYLLABUS		**3-1-0**
Unit	**Topic**	**Proposed Lecture**
I	Definition of fluid, Newton's law of viscosity, Units and dimensions-Properties of fluids, mass density, specific volume, specific gravity, viscosity, compressibility and surface tension, Incompressible flow, Bernoulli's equation and its applications - Pitot tube, orifice meter, and venturi meter.	**08**
II	Continuum & free molecular flows. Steady and unsteady, uniform and non-uniform, laminar and turbulent flows, rotational and irrotational flows, compressible and incompressible flows, subsonic, sonic and supersonic flows, sub-critical, critical and supercritical flows, one, two- and three dimensional flows, streamlines, continuity equation for 3D and 1D flows.	**08**
III	Equation of motion for laminar flow through pipes resistance to flow, minor losses, pipe in series and parallel, power transmission through a pipe, siphon, water hammer. Boundary layer thickness, boundary layer over a flat plate, laminar boundary layer, application of momentum equation, turbulent boundary layer, laminar sublayer, separation and its control, Drag and lift.	**08**
IV	Introduction to hydrodynamic thrust of jet on a fixed and moving surface, Classification of turbines, Impulse turbines, Constructional details, Velocity triangles, Power and efficiency calculations, Governing of Pelton wheel. Francis and Kaplan turbines, Constructional details, Velocity triangles, Power and efficiency Selection of water turbines.	**08**
V	Classifications of centrifugal pumps, Vector diagram, Work done by impellor, Efficiencies of centrifugal pumps, Specific speed, Cavitation & separation, reciprocating pump theory, Slip, Indicator diagram, Effect of acceleration, air vessels, Comparison of centrifugal and reciprocating pumps.	**08**

Books and References:

1. Introduction to fluid mechanics and Fluid machines by S.K Som, Gautam Biswas, S Chakraborty.
2. Fluid mechanics and machines by R.K Bansal.
3. F. M. White, Fluid Mechanics, 6th Ed., Tata McGraw-Hill, 2008.
4. Fluid Mechanics and Its Applications by V.K. Gupta et.al. 5. Fluid Mechanics by Yunus Cengel.
5. Batchelor, G. K. (1999). Introduction to fluid dynamics. New Delhi, India: Cambridge University Press.
6. Acheson, D. J. (1990). Elementary fluid dynamics. New York, USA: Oxford University Press.
7. R.W. Fox, A.T. McDonald and P.J. Pritchard, Introduction to Fluid Mechanics, 6th Ed., John Wiley, 2004.
8. Fluid Mechanics and Hydraulic Machines, 5/ed by Mahesh Kumar, Pearson Education
9. Fluid Mechanics, 2e in SI Units, by R.C Hibbeler, Pearson Education

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